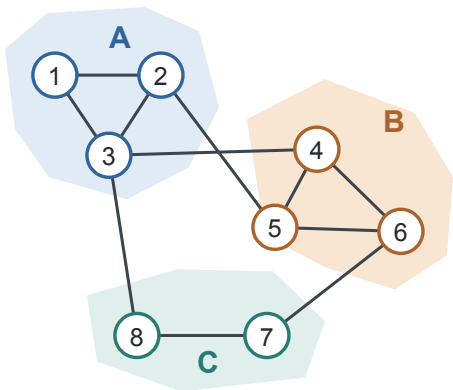


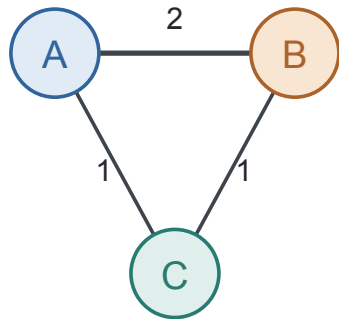
From nodes to a coarsened graph



(a) Original graph

	A	B	C
1	1	0	0
2	1	0	0
3	1	0	0
4	0	1	0
5	0	1	0
6	0	1	0
7	0	0	1
8	0	0	1

(b) Assignment S



(c) Coarsened graph

Pool features and edges

$$X' = S^T Z \quad A' = S^T A S$$

A: adjacency S: assignments Z: node embeddings

Illustrative hard assignments · DiffPool learns soft assignments

Pooled adjacency

Diagonal mass retained;
graph shows cross-cluster
edges.

	A	B	C
A	6	2	1
B	2	6	1
C	1	1	2